

#ASLI ENGINEERING

The Redis Object and Integer Internals

SWIPE

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The Redis Object

Every single object we put in Redis is wrapped in a Redis Object

↳ so that we keep some meta info

↳ manage object state & expiration

↳ make common features agnostic

```
struct redisObject {
    unsigned type: 4;
    unsigned encoding: 4;
    unsigned lru: LRU_BITS;
    int refCount;
    void * ptr;
};
```

1. Instead of assigning 4B to type & encoding Redis saves memory by allocating a few bits

   type → 4 bits
   encoding → 4 bits
   } 8 bits = 1B

2. void *ptr allows us to store / refer to any type of object
3. refCount → # places where same obj is referenced

   when refCount = 0 → free the object
4. lru takes up 24 bits → 3 Bytes

   to store information about time & freq of access

Total size : 4 + 4 + 24 + 32 + 32 → 12B
(4 + 4 + 24 = 4B; 32 = 4B; 32 = 4B)

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Type and Encoding

Redis supports 7 object types — Each object can be encoded differently

1. string ⟶ raw, int, embstr ← < 44 bytes
2. list ⟶ ziplist , linked list
3. set ⟶ intset, hashtable
4. sorted set ⟶ ziplist , skiplist
5. hash ⟶ ziplist , hashtable
6. module
7. stream

Ziplist is great at saving space

hence used when data is small

but when it grows beyond a threshold

it is converted to general type

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